Hospital Readmission Prediction using Machine Learning

Data 71200 Final Project

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Data Set

Source

Clore, J., Cios, K., DeShazo, J., & Strack, B. (2014). *Diabetes 130-US Hospitals for Years 1999-2008* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5230J.

Originally analyzed as part of Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., & Clore, J. N. (2014). Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records. *BioMed Research International, 2014*(1), 781670. https://doi.org/10.1155/2014/781670.

Context

- Anonymized data on hospital stays of 1-14 days for patients with diabetes (reason for admission may be different)
- Records from multiple EHR (Electronic Health Record) systems combined → inconsistent categories for some variables
- Multiple observations for some patients

Classification Target

Predict whether patient will be readmitted to hospital within 30 days of discharge

Data Set - Available Variables

Variable(s)	Type	Range / # Values	Description
readmitted Target	Categorical	3	“<30” if readmitted within 30 days, “>30” if readmitted after 30 days, “NO” if not readmitted
encounter_id	Integer	12522 - 443867222	Unique identifier for each observation
Patient Data			
patient_nbr	Integer	135 - 189502619	Unique identifier for each patient. There may be multiple observations in the data set corresponding to the same patient, if they had multiple hospitalizations meeting the inclusion criteria for the dataset.
race	Categorical	5	Options are Caucasian, African American, Hispanic, Asian, other. May also be missing (unknown race).
age	Categorical	10	Provided in 10-year bins e.g. [20, 30)
gender	Categorical	3	Male, female, unknown/other
weight	Categorical	10	Provided in 25-kg bins e.g. [50, 75). Frequently missing.
payer_code	Categorical	23	Insurance type. Frequently missing.

Data Set - Available Variables

Hospital Encounter Data			
admission_type_id	Categorical	9	Category of hospital admission e.g. emergency, elective
admission_source_id	Categorical	21	How the patient came to be hospitalized, e.g. transfer from another hospital, referral by primary care provider, emergency room
discharge_disposition_id	Categorical	29	Where the patient was sent after being discharged, e.g. home, long-term care
time_in_hospital	Integer	1 - 14	Days spent in hospital. The original research which resulted in this dataset excluded patients who were hospitalized more than two weeks or less than one day.
medical_specialty	Categorical	84	Specialty of doctor admitting the patient to hospital, e.g. internal medicine, cardiology. Frequently missing.
diag_1, diag_2, diag_3	String	~1000	ICD-9 diagnostic codes corresponding to up to primary diagnosis for which the patient was admitted and up to two secondary diagnoses. While all patients in the sample were diabetic, this was not necessarily the primary reason they were hospitalized.
number_diagnoses	Integer	1 - 16	Total number of diagnoses in the patient’s chart (not necessarily all related to current hospitalization).

Data Set - Available Variables

Tests and Medications			
num_lab_procedures	Integer	1 - 129	Number of lab tests run during hospitalization
num_procedures	Integer	0 - 6	Number of other procedures (e.g. scans, surgery) performed
max_glu_serum	Categorical	4	“None” if serum blood glucose was not measured; “normal”, “>200”, or “>300” depending on test result if it was measured.
A1Cresult	Categorical	4	“None” if A1C (long-term blood glucose) was not measured; “normal”, “>6”, or “>7” depending on test result if it was measured
num_medications	Integer	1 - 81	Number of distinct medications administered
diabetesMed	Boolean	-	Whether the patient was taking any medication for diabetes
change	Boolean	-	Whether any new medications were prescribed OR dosage of current medications was adjusted
[24 specific diabetes medication variables]	Categorical	4	“None” if medication not prescribed, “Steady” if prescribed and dosage unchanged, “Up” or “Down” if dosage changed
Recent Medical History			
number_outpatient	Integer	0 - 40	Number of times the patient was seen in an outpatient setting in the year leading up to the hospitalization
number_inpatient	Integer	0 - 19	As above but for inpatient setting
number_emergency	Integer	0 - 54	As above but for emergency setting

Data Cleaning

Remove invalid data

- Discharge reasons included “Expired” and “discharged to hospice” – does not make sense to predict readmission for these patients

Limit to one row per patient

- Some patient IDs have multiple associated encounters → potential non-independence
- Keep last encounter for each patient, but use previous encounters to create new features so less information is lost
    - # distinct diagnoses
    - # previous admissions
    - # previous readmissions within 30 days
    - Total days spent in hospital

Handle missing values

- Almost all values missing → drop variable
    - Weight
    - Some medication variables
- Fewer missing values → create “Unknown” category
    - Race
    - Medical specialty of referrer
    - Blood glucose (high missingness, but relevant per original paper)
    - Payer code (insurance)

Data Cleaning

Convert ID fields to values; combine redundant categories

- Admission type ID
- Discharge disposition ID
- Admission source ID

admission_type_id	description
1	Emergency
2	Urgent
3	Elective
4	Newborn
5	Not Available
6	NULL
7	Trauma Center
8	Not Mapped

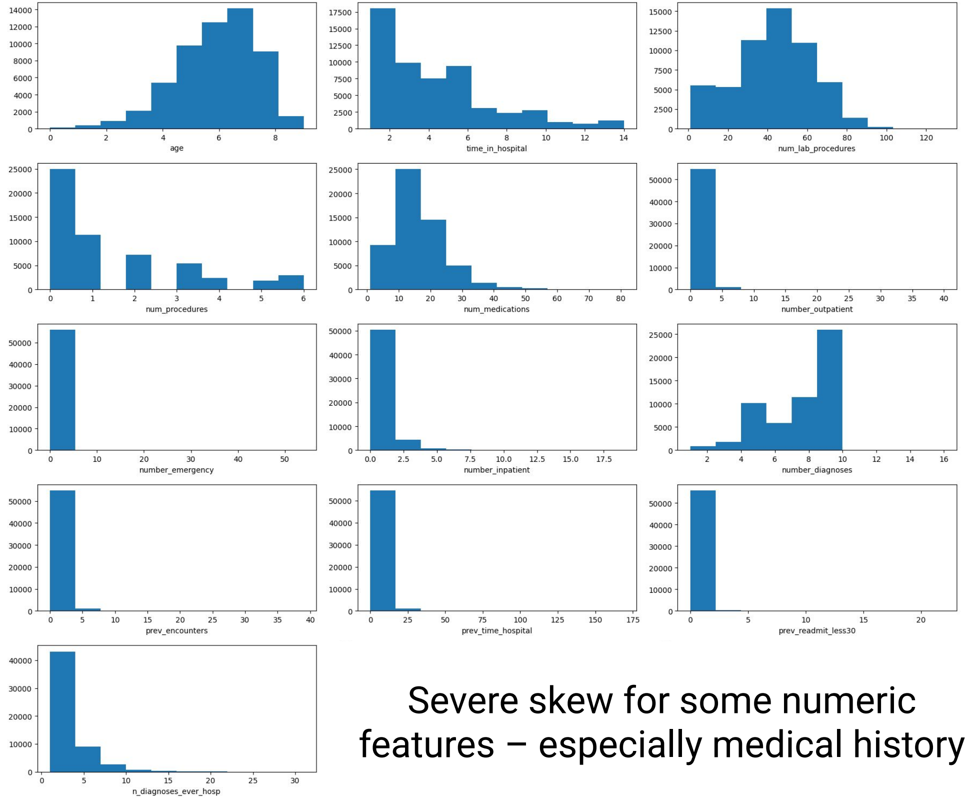
Unknown

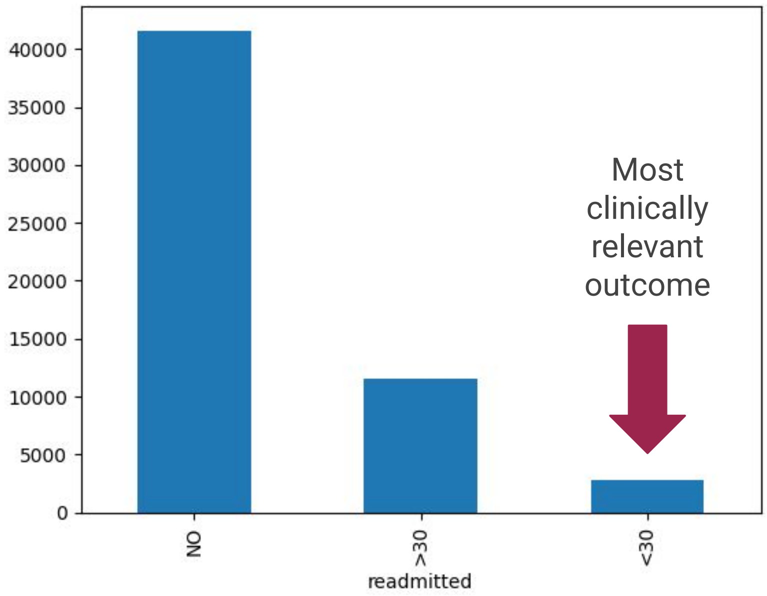
- Medical specialty already categorical, but categories needed to be collapsed

Convert ICD-9 codes to primary diagnosis categories

broad_category	
DISEASES OF THE CIRCULATORY SYSTEM	16667
ENDOCRINE, NUTRITIONAL AND METABOLIC DISEASES, AND IMMUNITY DISORDERS	6095
DISEASES OF THE RESPIRATORY SYSTEM	5207
DISEASES OF THE DIGESTIVE SYSTEM	5124
SYMPTOMS, SIGNS, AND ILL-DEFINED CONDITIONS	4337
INJURY AND POISONING	3920
DISEASES OF THE MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE	3096
DISEASES OF THE GENITOURINARY SYSTEM	2821
NEOPLASMS	1998
INFECTIOUS AND PARASITIC DISEASES	1388
DISEASES OF THE SKIN AND SUBCUTANEOUS TISSUE	1352
MENTAL DISORDERS	1287
Other	952
DISEASES OF THE NERVOUS SYSTEM AND SENSE ORGANS	667
DISEASES OF THE BLOOD AND BLOOD-FORMING ORGANS	544
COMPLICATIONS OF PREGNANCY, CHILDBIRTH, AND THE PUERPERIUM	473
CONGENITAL ANOMALIES	32

Variable Distributions

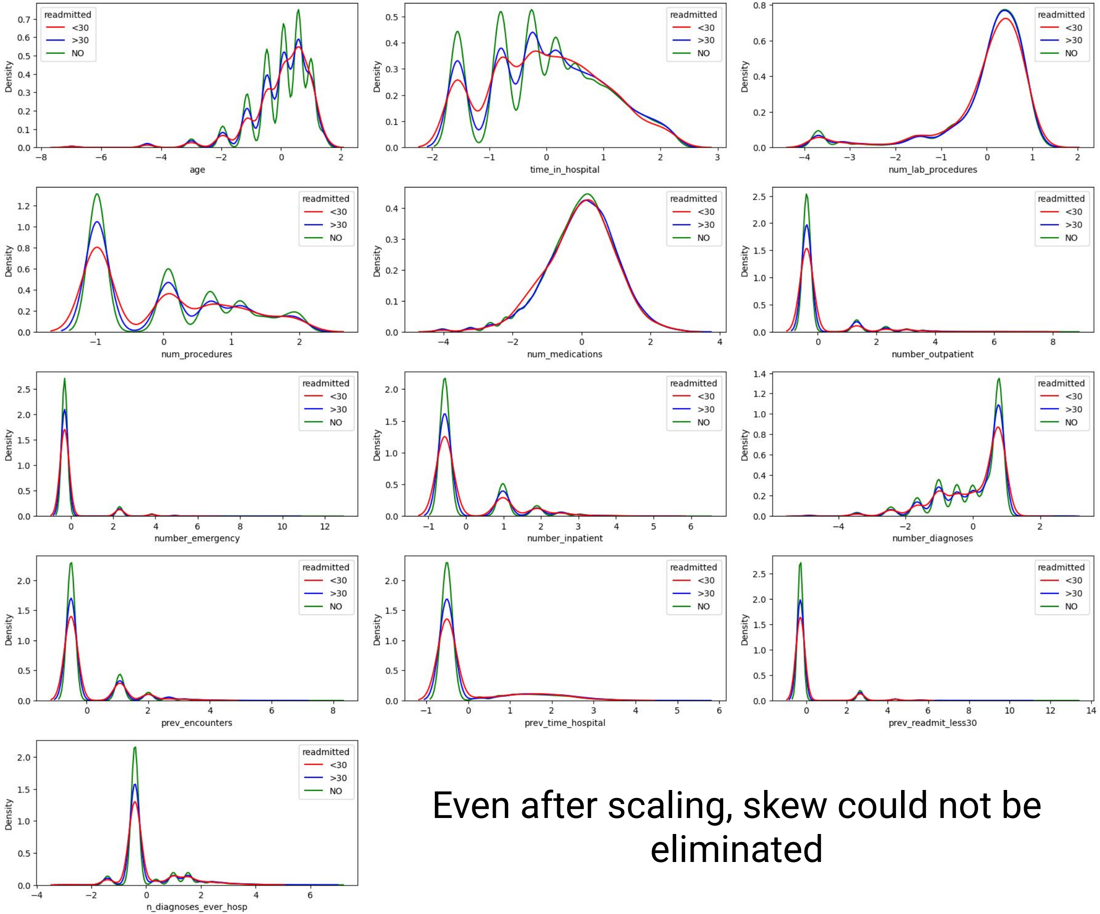




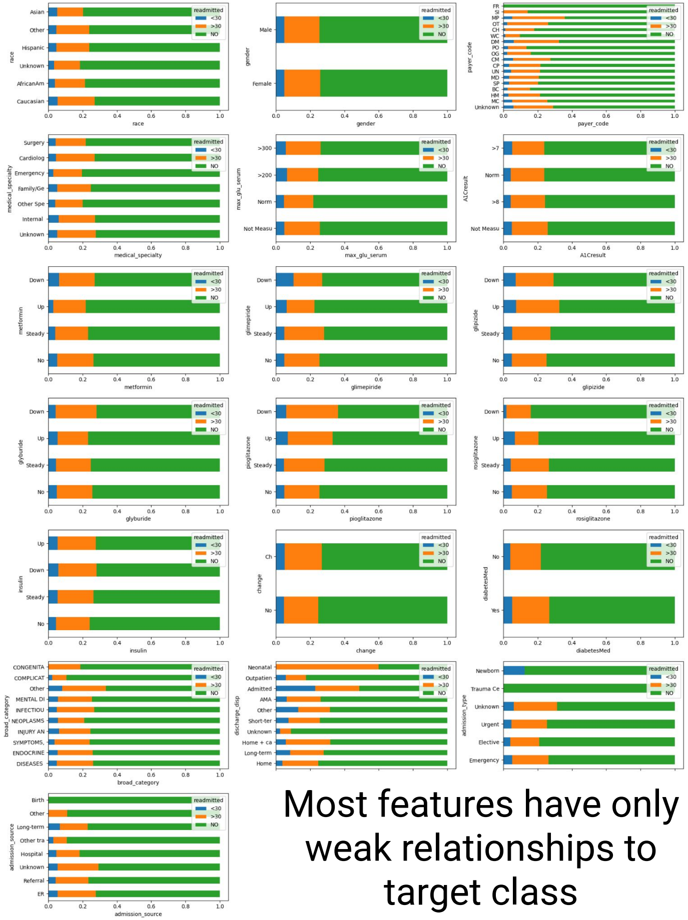
Severe skew for some numeric features – especially medical history

Imbalance in target feature

Variable Distributions



Even after scaling, skew could not be eliminated



Most features have only weak relationships to target class

Supervised Learning - Multiclass

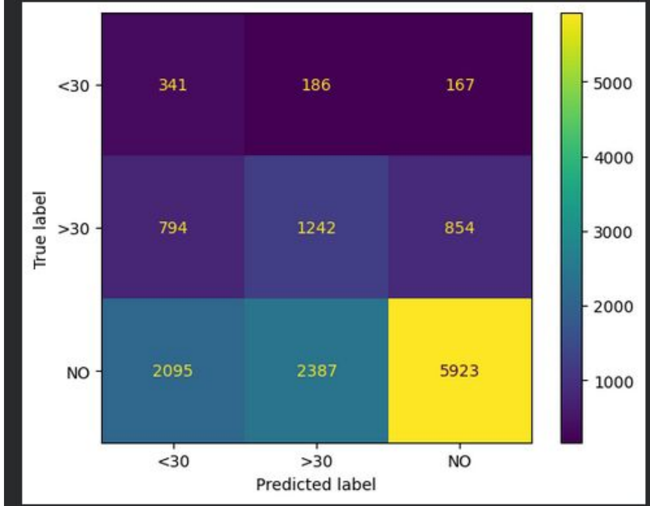
Logistic Regression

```
Classification report:

              precision    recall  f1-score   support

         <30       0.11      0.49      0.17       694
         >30       0.33      0.43      0.37      2890
          NO       0.85      0.57      0.68     10405

    accuracy                           0.54     13989
   macro avg       0.43      0.50      0.41     13989
weighted avg       0.71      0.54      0.59     13989
```



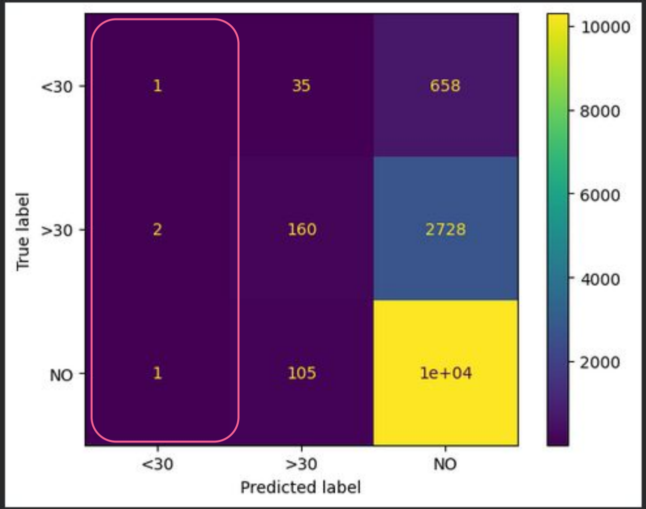
Random Forest

```
Classification report:

              precision    recall  f1-score   support

         <30       0.25      0.00      0.00       694
         >30       0.53      0.06      0.10      2890
          NO       0.75      0.99      0.86     10405

    accuracy                           0.75     13989
   macro avg       0.51      0.35      0.32     13989
weighted avg       0.68      0.75      0.66     13989
```

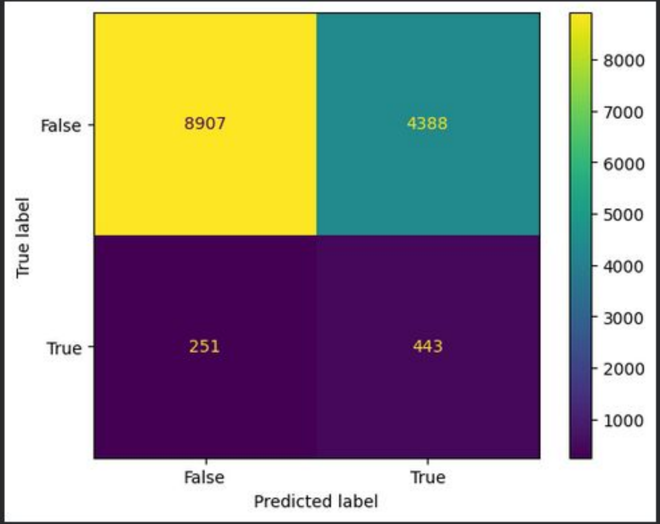


Supervised Learning - Binary

Logistic Regression

Classification report:

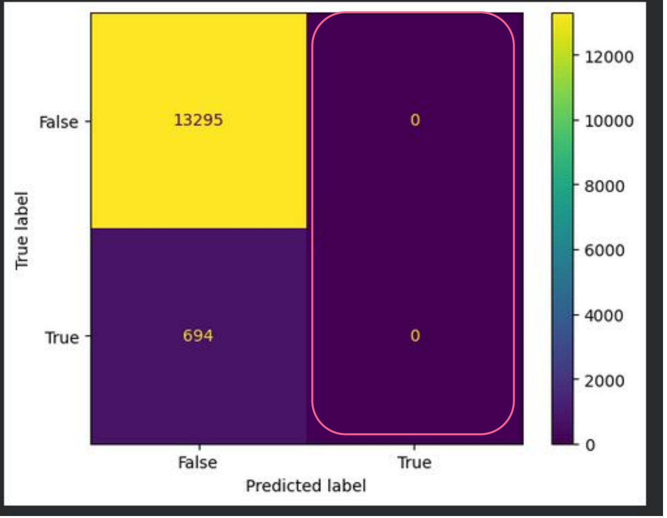
	precision	recall	f1-score	support
False	0.97	0.67	0.79	13295
True	0.09	0.64	0.16	694
accuracy			0.67	13989
macro avg	0.53	0.65	0.48	13989
weighted avg	0.93	0.67	0.76	13989



Random Forest

Classification report:

	precision	recall	f1-score	support
False	0.95	1.00	0.97	13295
True	0.00	0.00	0.00	694
accuracy			0.95	13989
macro avg	0.48	0.50	0.49	13989
weighted avg	0.90	0.95	0.93	13989



Supervised Learning - Grid Search

Logistic Regression

- **C** (regularization parameter)
  - 0.01, 0.1, 1, 10, 100
- **l1_ratio** (Lasso, Ridge, or mixture of both)
  - 0, 0.25, 0.5, 0.75, 1
  - 0 corresponds to Ridge (L2) only
  - 1 corresponds to Lasso (L1) only

Random Forest

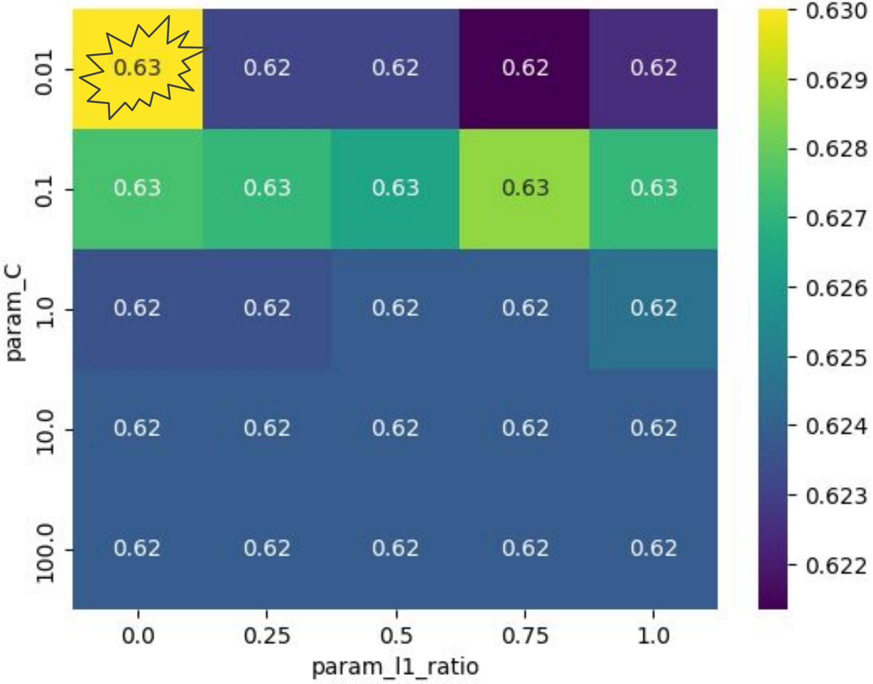
- **max_depth**
  - 5, 10, 15, 20, 25
- **min_weight_fraction_leaf**
  - 0, 0.005, 0.01, 0.05, 0.1
  - When fitting random forest with class_weight = 'balanced' or 'balanced_subsample', observations are weighted based on class prevalence
- And many more…but none improved performance

Custom scorer to optimize for high recall on specific class (in this case True for readmission within 30 days) – default recall_score would prioritize majority class

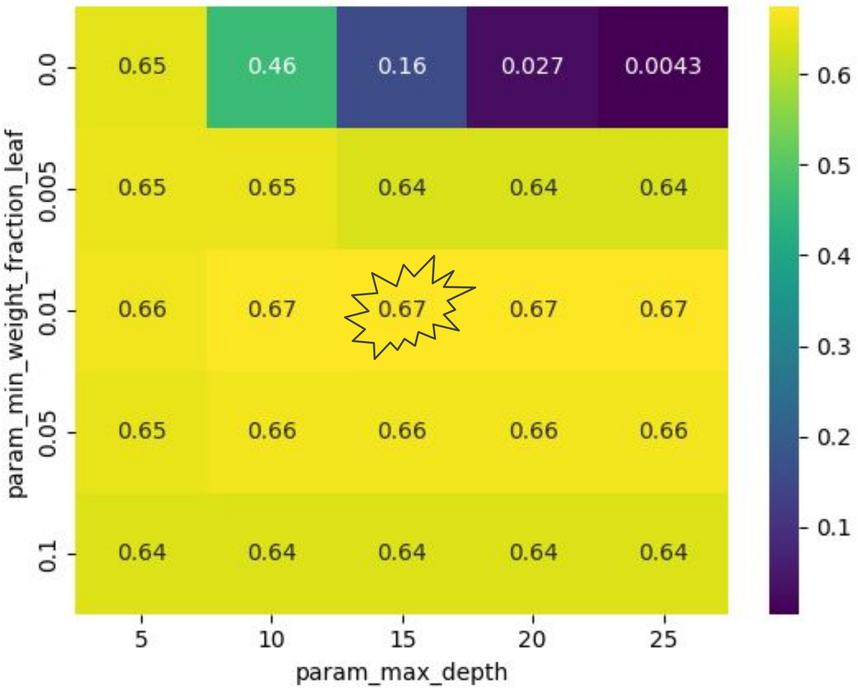
```
GridSearchCV(model, cv=kfold, scoring = make_scorer(recall_score, average=None,
labels=[pos], pos_label=pos), param_grid=param_grid, error_score='raise', verbose=0)
```

Supervised Learning - Grid Search

Logistic Regression - very little variation



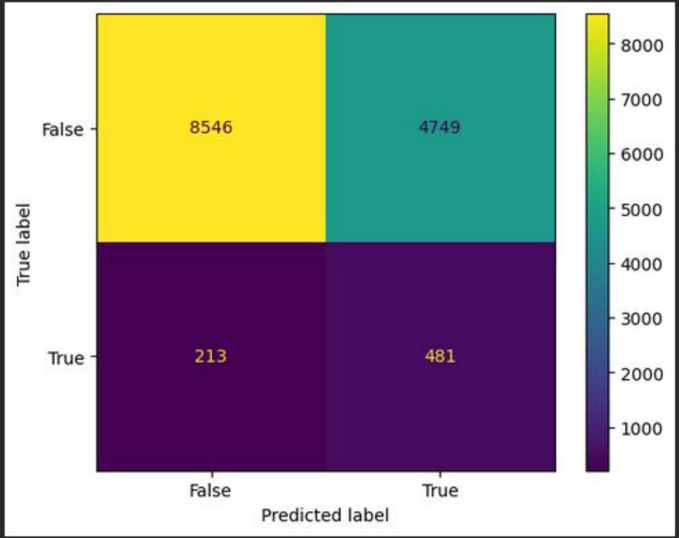
Random Forest - max_depth had minimal effect compared to minimum leaf weight



Final Random Forest model - performance on test set

Classification report:

	precision	recall	f1-score	support
False	0.98	0.64	0.78	13295
True	0.09	$\boxed{0.69}$	0.16	694
accuracy			0.65	13989
macro avg	0.53	0.67	0.47	13989
weighted avg	0.93	0.65	0.74	13989



Still Bad!

Definitely not acceptable performance for a clinical setting

Unsupervised Learning

Because many clustering algorithms require distances between all pairs of points to be calculated, Colab memory limitations were an issue.

Even using approximation options (precalculating connectivity, ball-tree distance), couldn't fit models

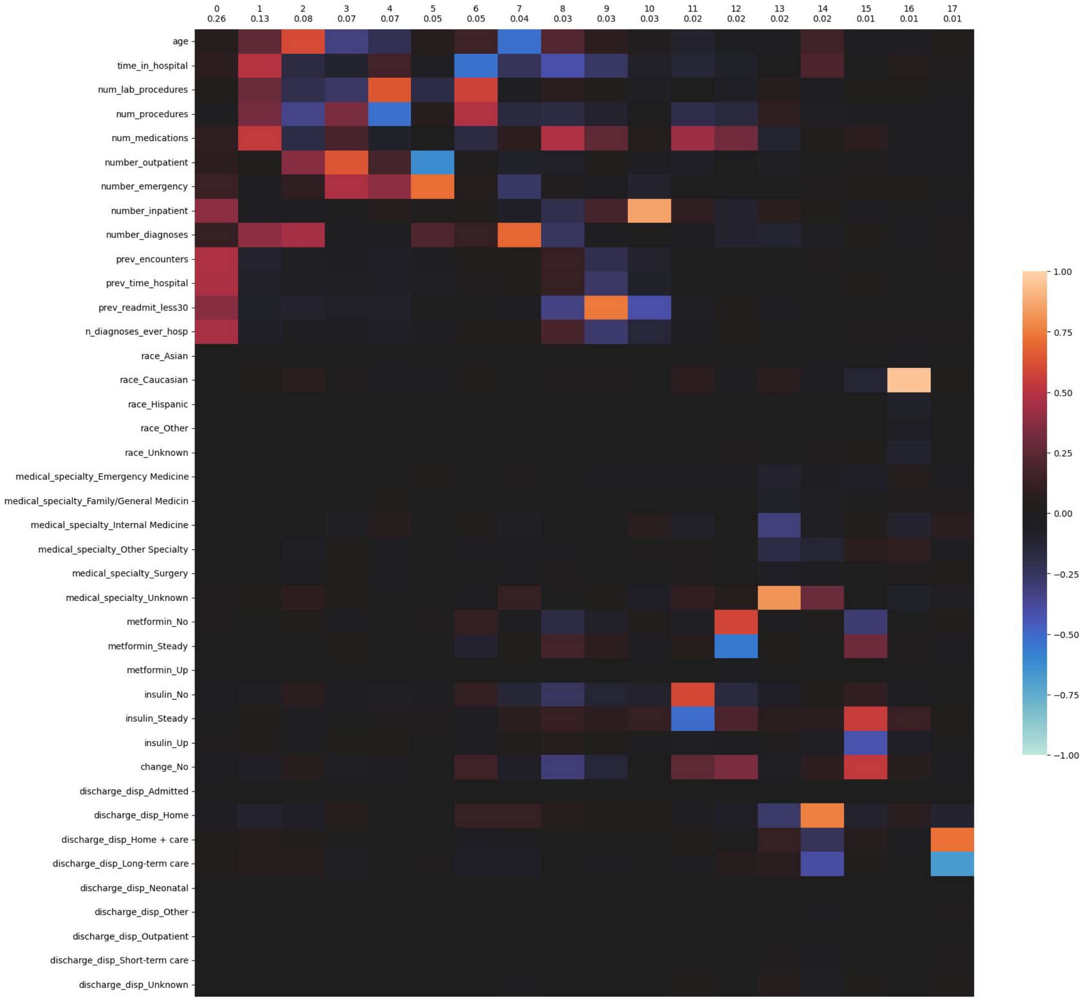
- Removed some categorical variables which were less associated with target
  - Categorical variables kept: race, medical specialty, metformin, insulin, discharge disposition
- Took (stratified) sample of half the rows in training set
  - Final sample for clustering: 27,979 observations, 40 features
- RF performance slightly lower, but comparable

Classification report:

	precision	recall	f1-score	support
False	0.97	0.66	0.79	13295
True	0.09	$\boxed{0.66}$	0.16	694
accuracy			0.66	13989
macro avg	0.53	0.66	0.47	13989
weighted avg	0.93	0.66	0.76	13989

PCA

- 18 components needed to explain 95% of variance
- Highest loading factors were based on numeric variables
- Medical history-related variables I constructed loaded on the first component, which explained 26% of variance
- Many variables uncorrelated or only weakly correlated with others

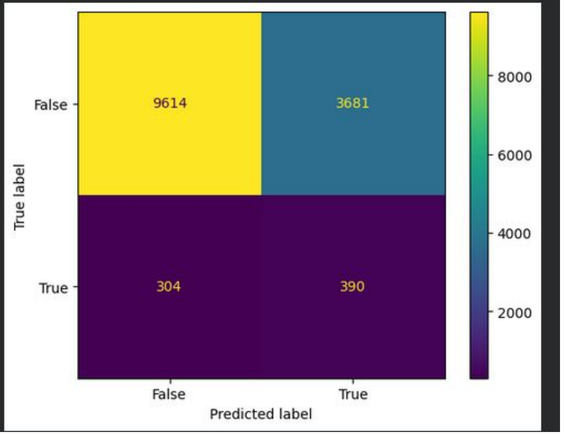


Model performance using PCA features

Previous best hyperparameter combination (max_depth 15, min_weight_fraction_leaf 0.01) was now overfitting

Classification report:

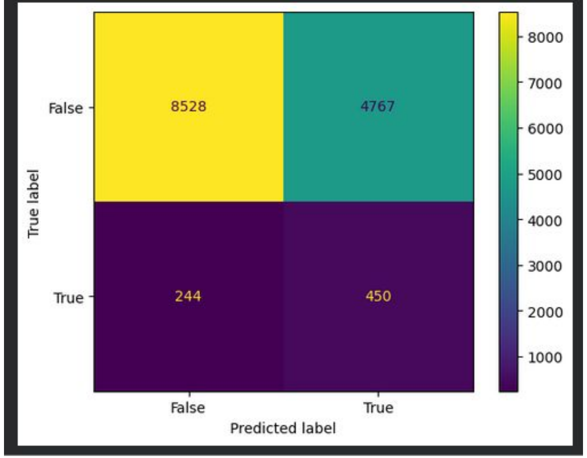
	precision	recall	f1-score	support
False	0.97	0.72	0.83	13295
True	0.10	0.56	0.16	694
accuracy			0.72	13989
macro avg	0.53	0.64	0.50	13989
weighted avg	0.93	0.72	0.80	13989



Increasing pruning by raising min_weight_fraction_leaf improved performance, but best setting (0.035) did not outperform original model

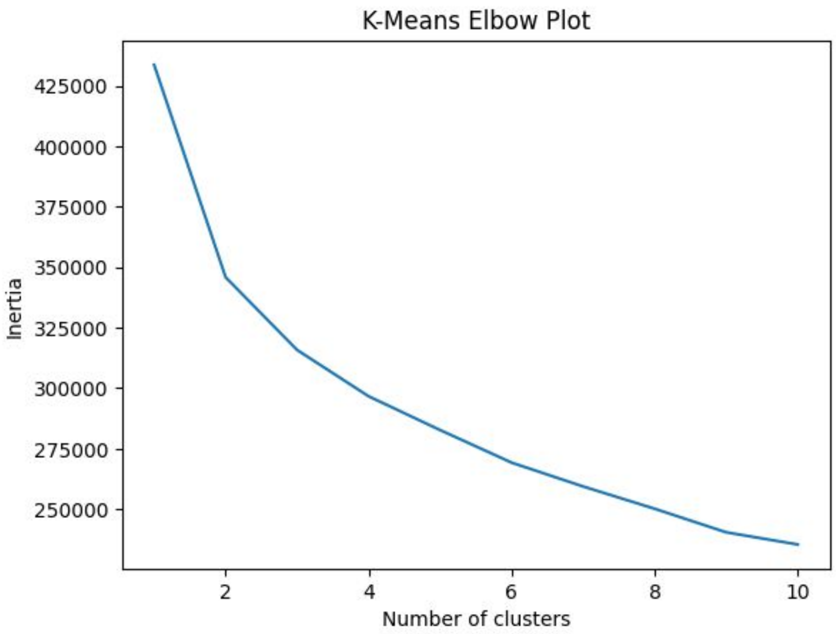
Classification report:

	precision	recall	f1-score	support
False	0.97	0.64	0.77	13295
True	0.09	0.65	0.15	694
accuracy			0.64	13989
macro avg	0.53	0.64	0.46	13989
weighted avg	0.93	0.64	0.74	13989

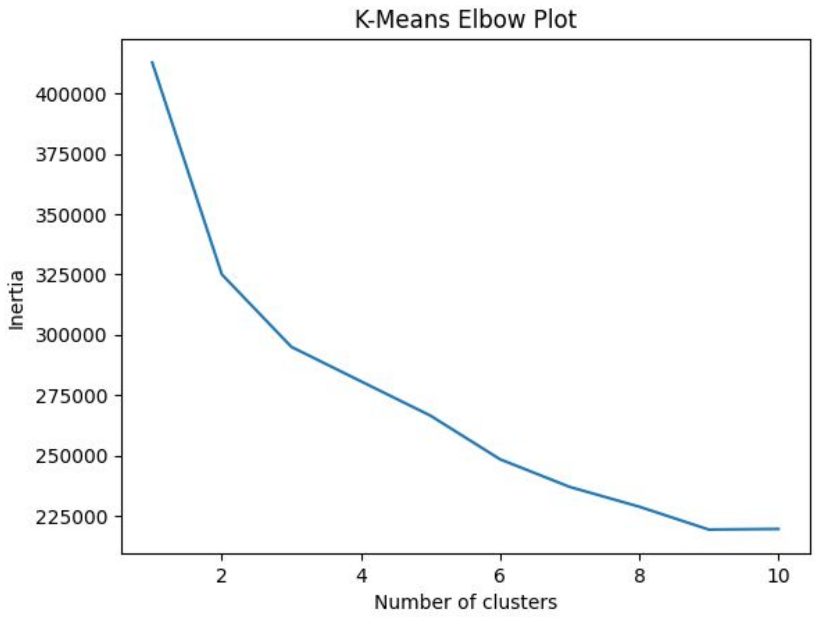


K-Means Clustering

Without PCA



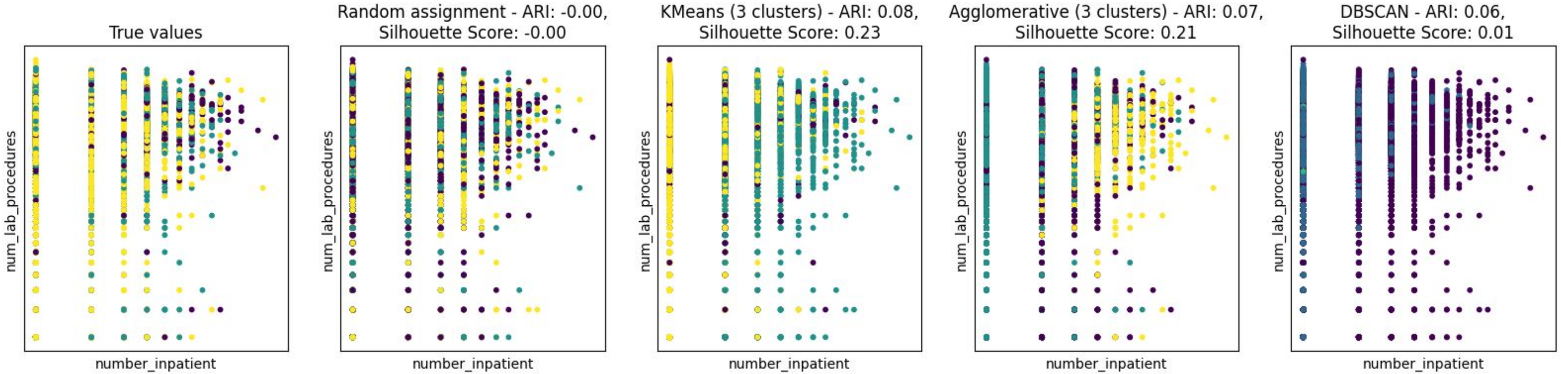
With PCA



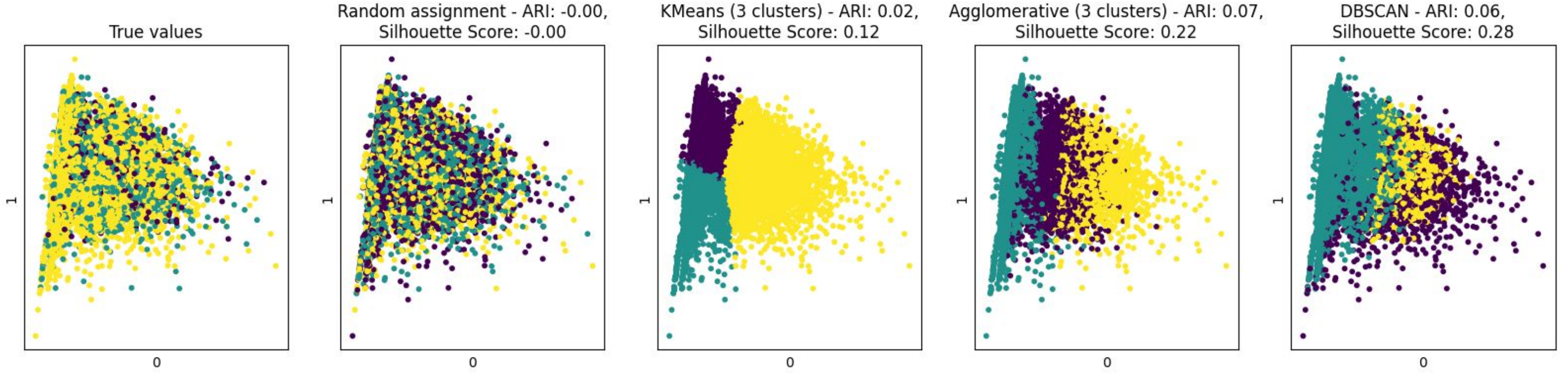
Elbow not clear…two or three clusters plausible based on target values

Clustering - 3 clusters

Without PCA

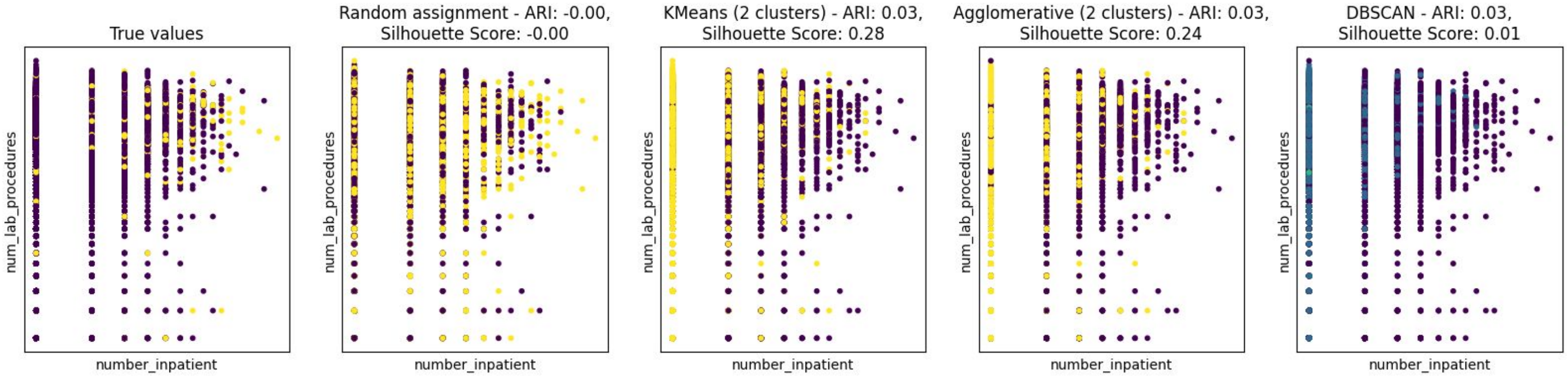


With PCA

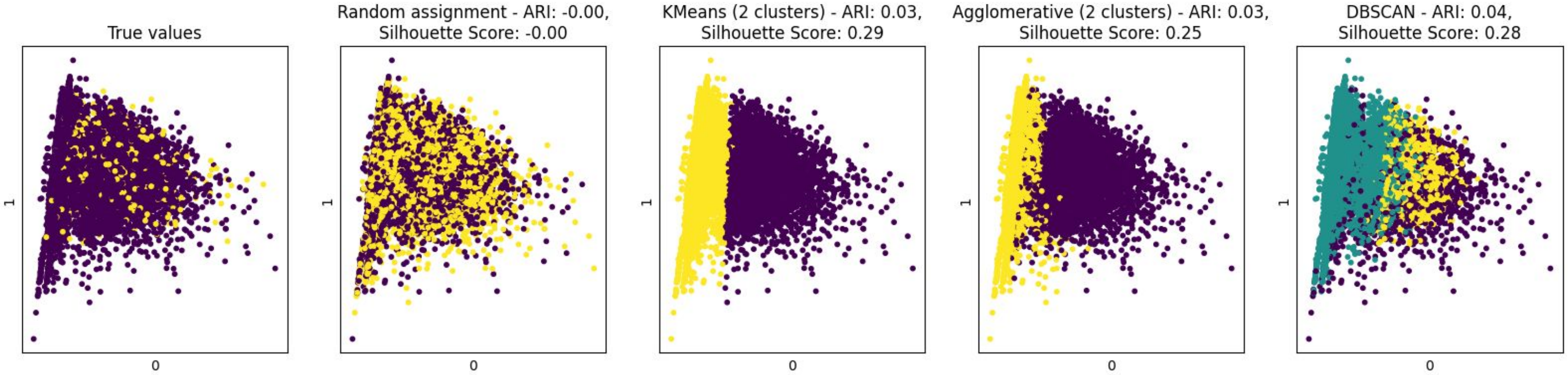


Clustering - 2 clusters

Without PCA



With PCA



Questions?